

Aim: To design, simulate, and verify the functionality of Half Adder, Full Adder, and 4-bit Adder circuits using fundamental logic gates, and to implement Half Adder and Full Adder circuits exclusively with NAND gates.

Theory:

1. Half Adder (Using Basic Logic Gates)

A Half Adder is a digital combinational circuit designed to perform the addition of two one-bit binary inputs, A and B. It generates two outputs:

- Sum (S): Represents the least significant bit of the addition
- Carry (C): Represents the overflow bit
 - Boolean Expressions: **Sum:** $S = A \oplus B$. And **Carry:** $C = A \cdot B$

The XOR gate is used to produce the Sum output because it gives a logic HIGH when the two inputs differ. The AND gate generates the Carry output only when both inputs are HIGH.

2. Half Adder (Using NAND Gates Only)

Since NAND gates are universal gates, any digital logic function can be implemented using only NAND gates.

- The XOR operation required for Sum is constructed by combining multiple NAND gates
- The AND operation for Carry is also derived from NAND gate configurations

Thus, a complete Half Adder circuit can be designed without using XOR or AND gates directly.

3. Full Adder (Using Basic Logic Gates)

A Full Adder is a combinational circuit that adds three binary inputs:

- A , B , Carry-in (cin)

It produces:

- Sum (S)
- Carry-out (Cout)

Boolean Expressions:

- Sum: $S = A \oplus B \oplus C_{in}$
- Carry: $C_{out} = AB + C_{in}(A \oplus B)$

The Full Adder can be realized by connecting two Half Adders in sequence and combining their carry outputs through an OR gate.

4. 4-bit Adder (Using Basic Logic Gates)

A 4-bit Adder is used for adding two 4-bit binary numbers. It is formed by cascading four Full Adders in series.

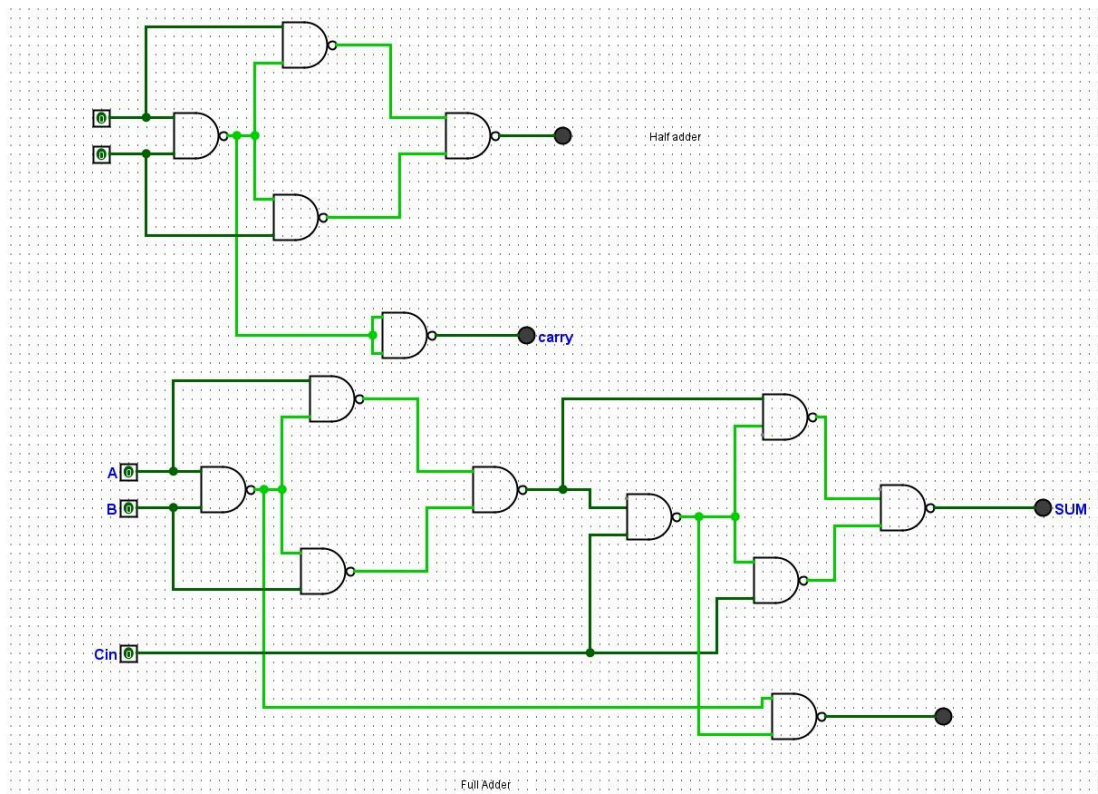
Working Principle:

- Each Full Adder handles one bit position
- The carry generated from each stage is forwarded to the next stage
- The first Full Adder receives $C_{in} = 0$
- The final output consists of:
 - A 4-bit Sum
 - A final Carry-out

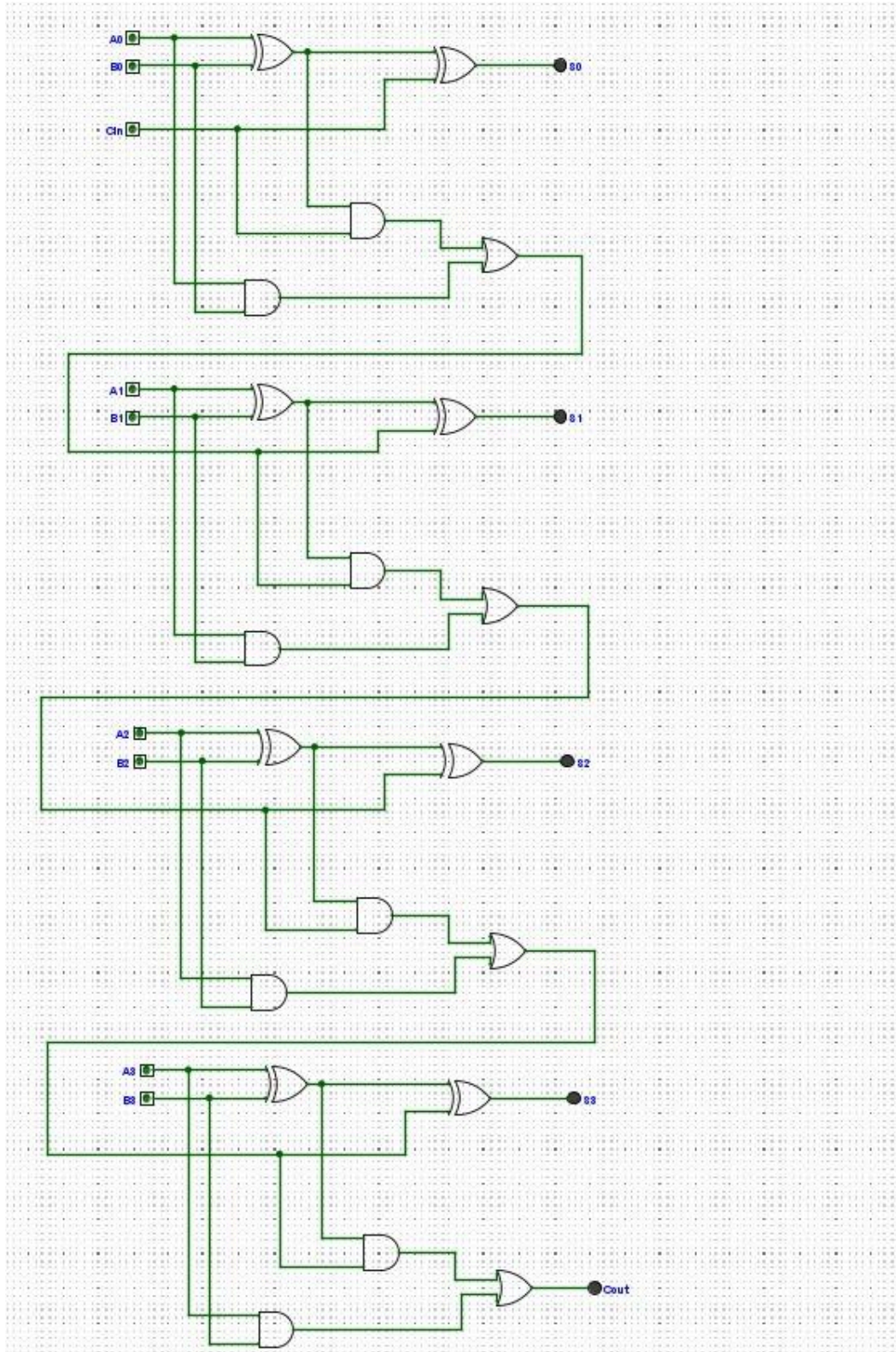
This arrangement is commonly referred to as a Ripple Carry Adder due to sequential carry propagation.

Logisim Implementation:

Circuit diagrams for Half Adder, Full Adder, and 4-bit Adder were designed and tested in Logisim to verify correct logical operation.



Full adder and half adder



4 bit full adder